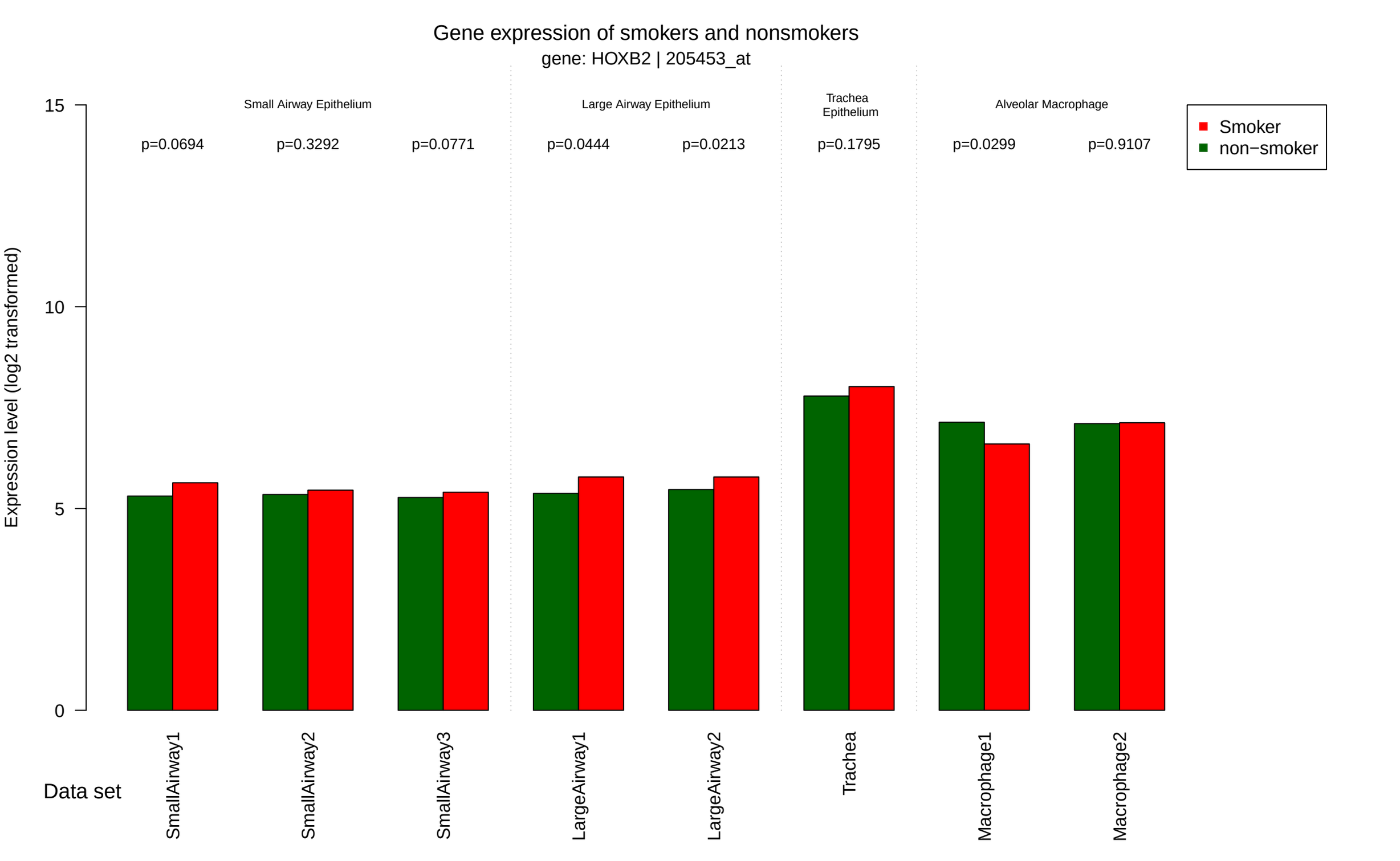
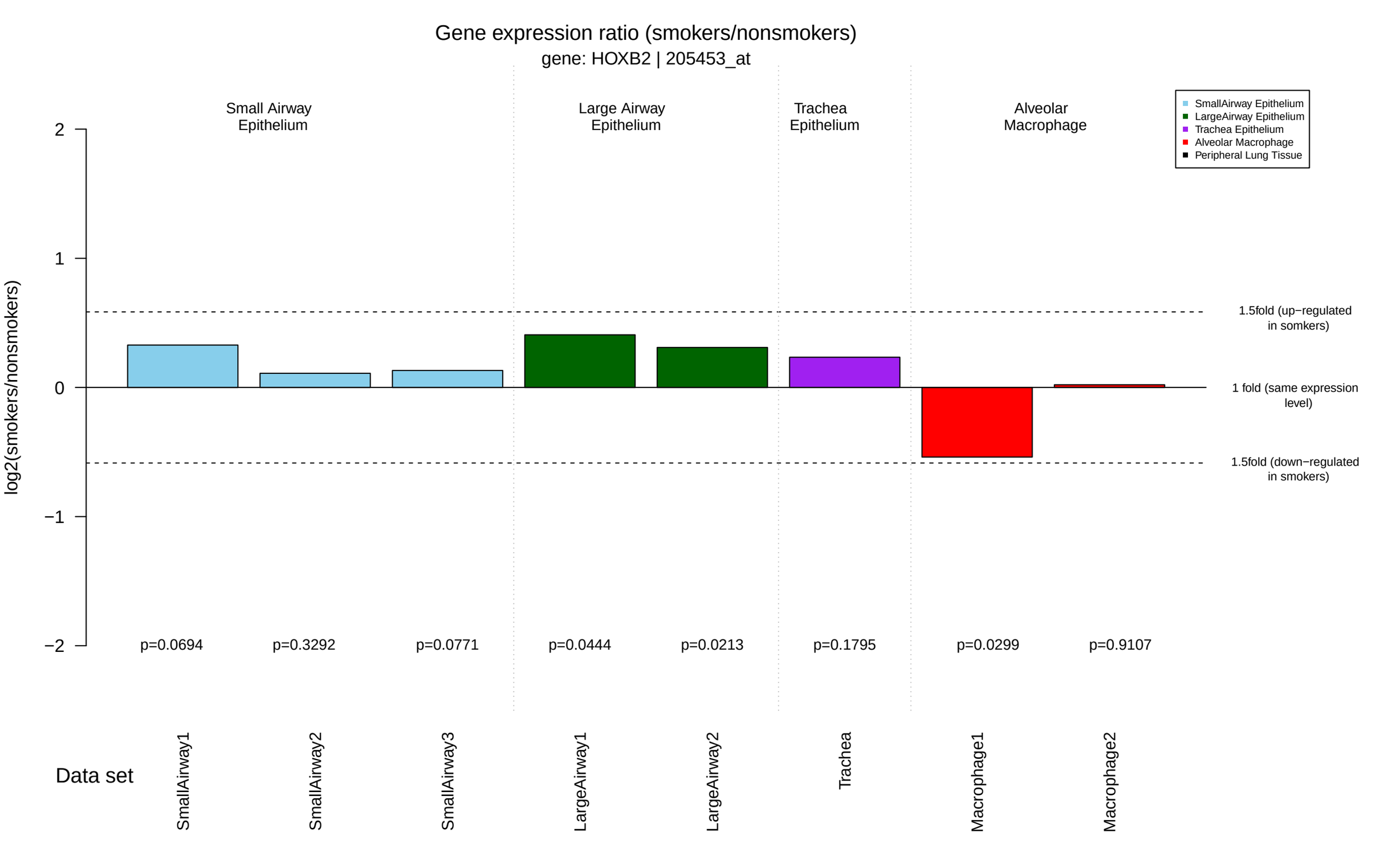
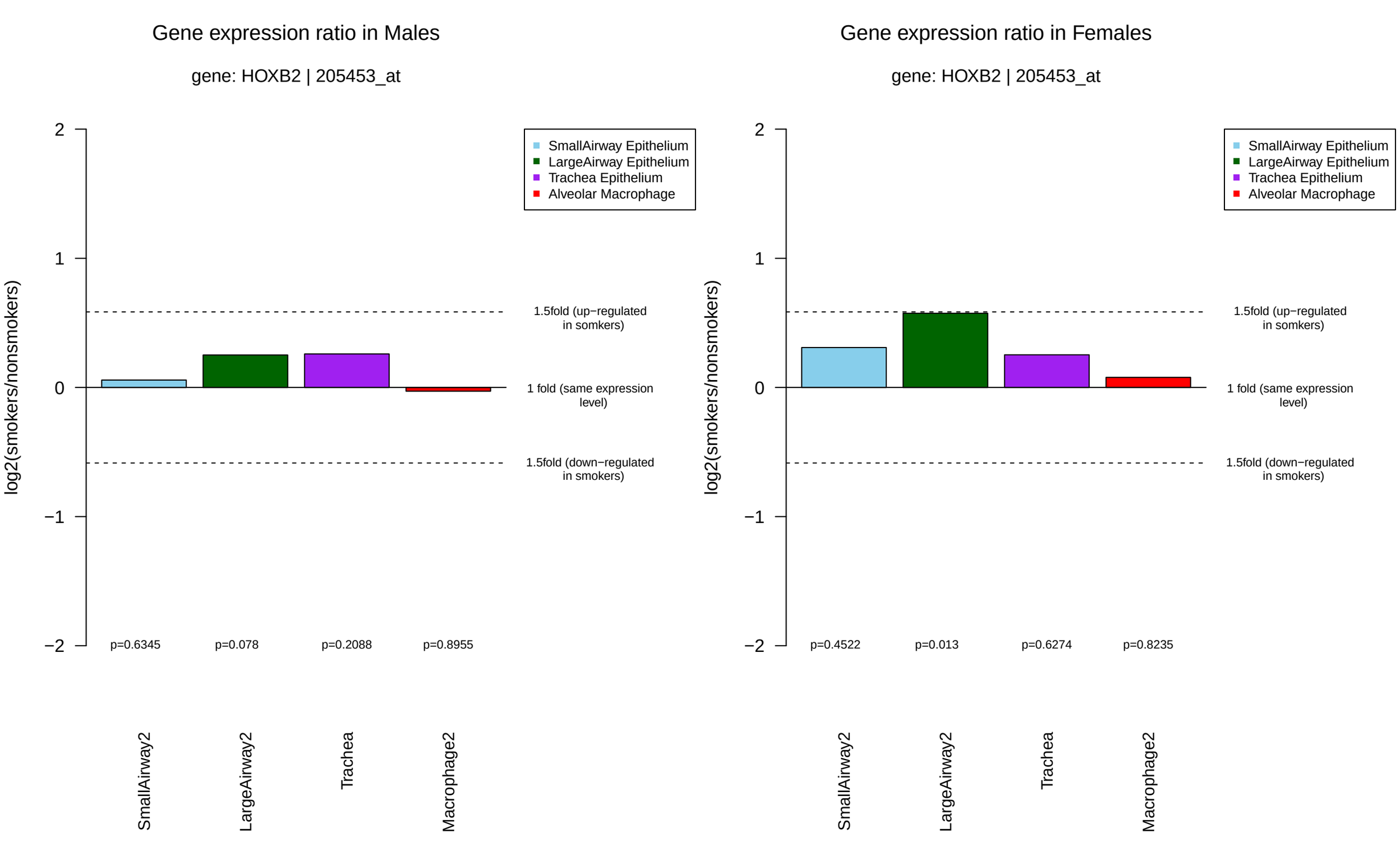
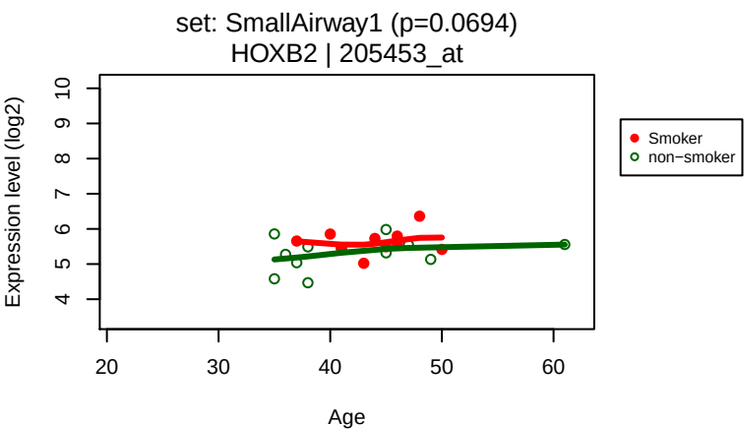
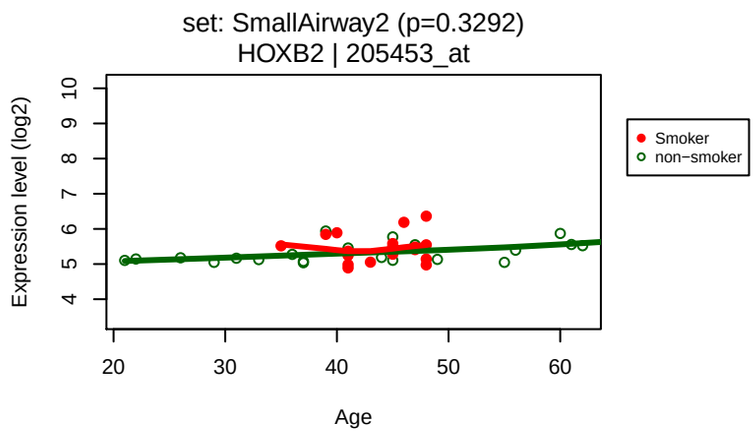
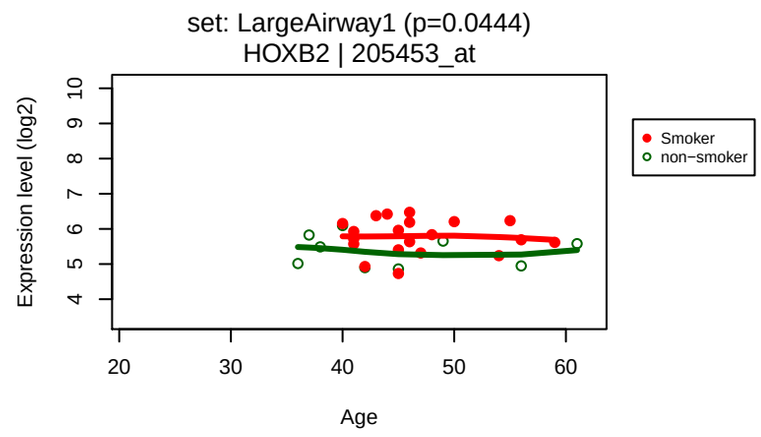

Gene expression of smokers and nonsmokers
gene: HOXB2 | 205453_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0694
p=0.3292
p=0.0771
p=0.0444
p=0.0213
p=0.1795
p=0.0299
p=0.9107
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

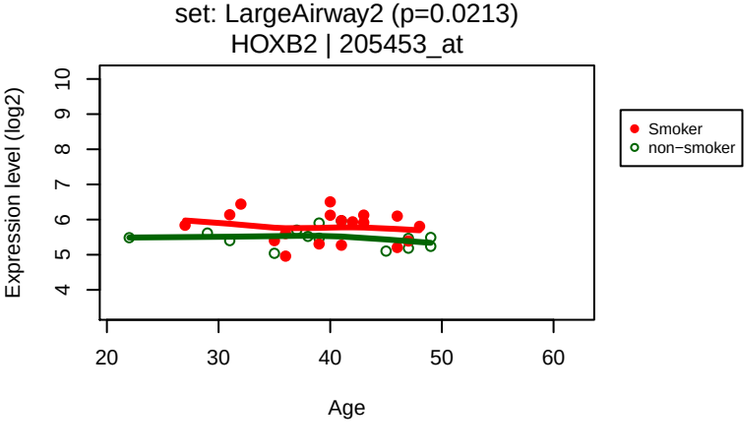

Gene expression ratio (smokers/nonsmokers)
gene: HOXB2 | 205453_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0694
p=0.3292
p=0.0771
p=0.0444
p=0.0213
p=0.1795
p=0.0299
p=0.9107
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

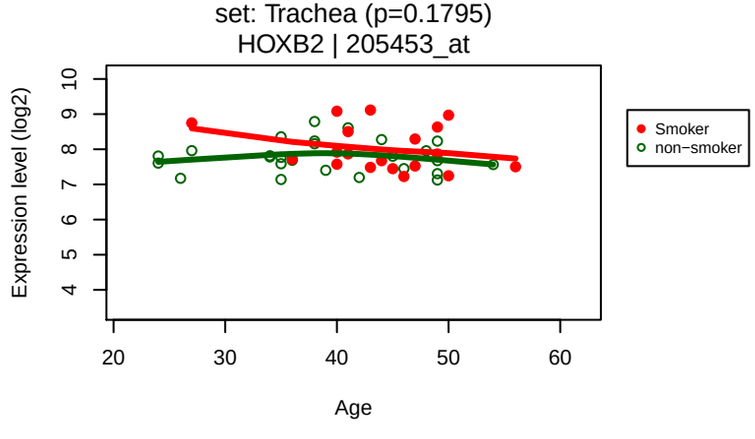

Gene expression ratio in Males
gene: HOXB2 | 205453_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6345
p=0.078
p=0.2088
p=0.8955
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HOXB2 | 205453_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4522
p=0.013
p=0.6274
p=0.8235
SmallAirway2
LargeAirway2
Trachea
Macrophage2

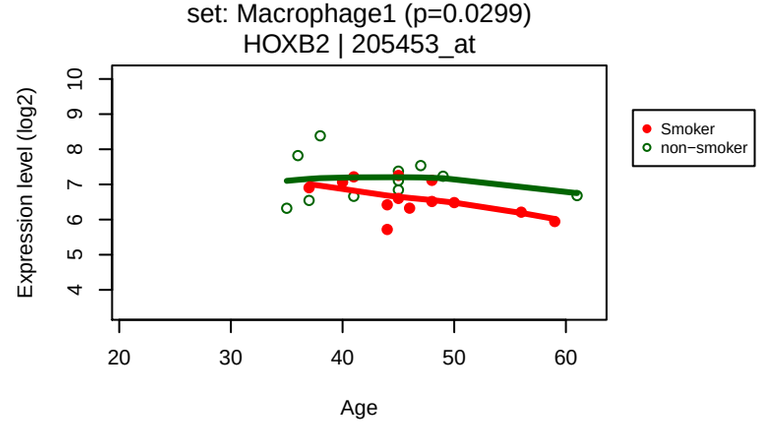

set: SmallAirway1 (p=0.0694)
HOXB2 | 205453_at
Expression level (log2)
Age
Smoker
non−smoker

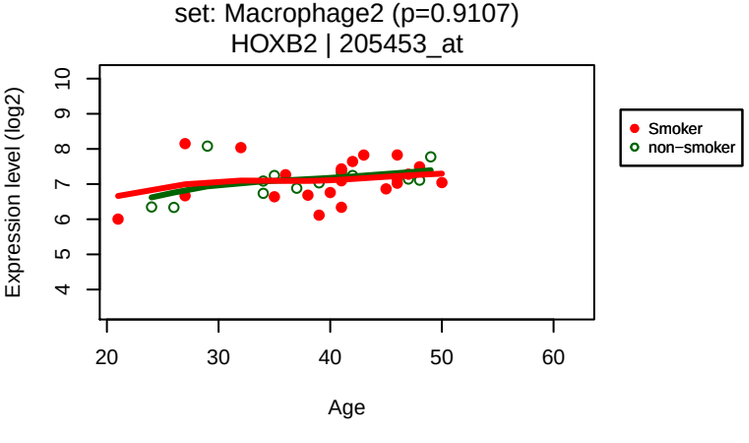

set: SmallAirway2 (p=0.3292)
HOXB2 | 205453_at
Expression level (log2)
Age
Smoker
non−smoker

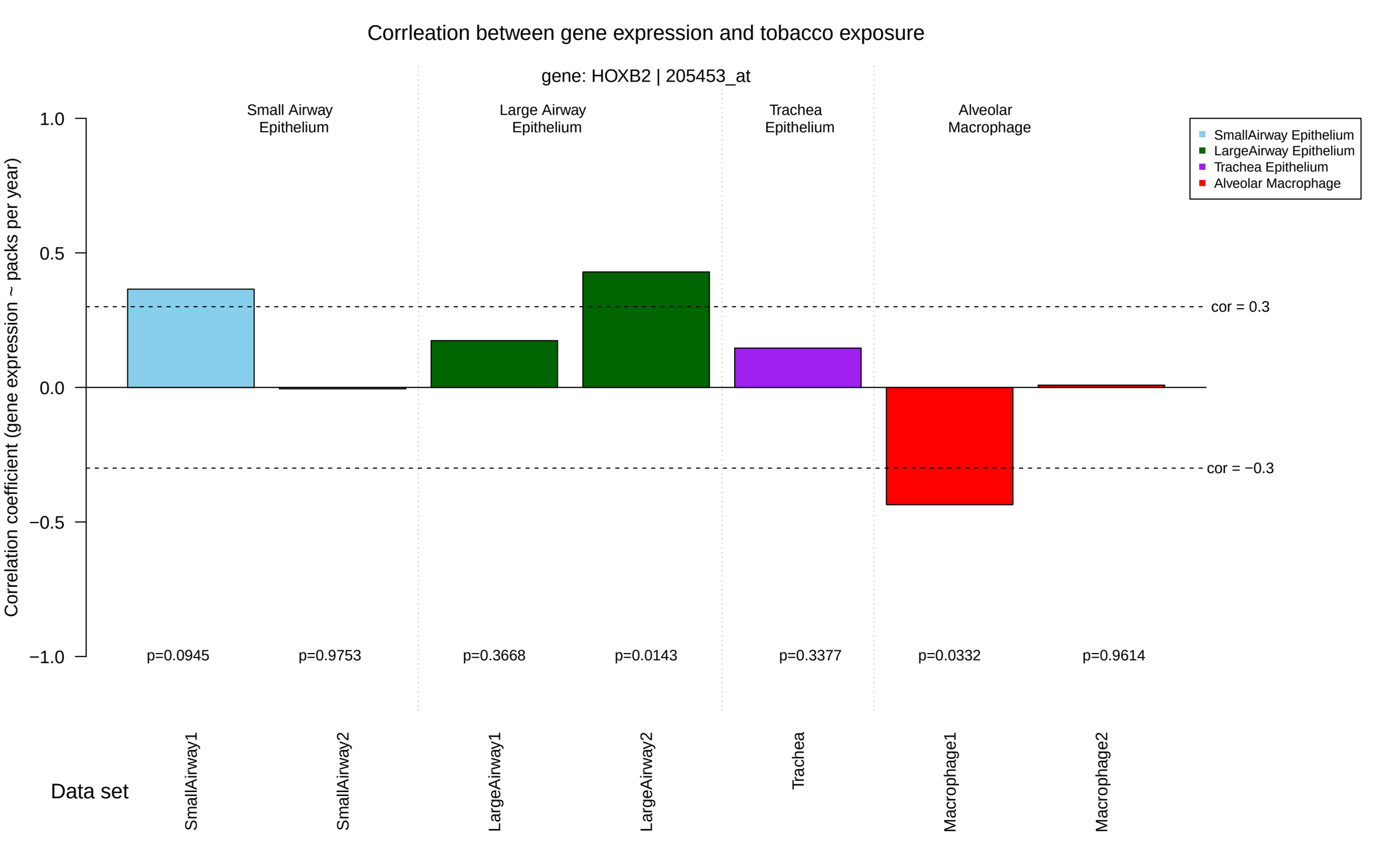

set: LargeAirway1 (p=0.0444)
HOXB2 | 205453_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0213)
HOXB2 | 205453_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1795)
HOXB2 | 205453_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0299)
HOXB2 | 205453_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9107)
HOXB2 | 205453_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HOXB2 | 205453_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0945
p=0.9753
p=0.3668
p=0.0143
p=0.3377
p=0.0332
p=0.9614
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2